

Real Time Feature Point Detectors for Determining Ego Motions

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Abstract

Ego motion determination is an important application to recognize the motion of an observer in real time. Here we have implemented feature point detector using FFME for ego-motion determination. The main objective of this work is to detect the motions of any objects such as humans, animals, vehicles, missiles and so on. The camera plays a vital role in surveillance of isolated and or restricted area. To improve the intelligence of camera some efficient video processing algorithms are needed. In this process, the detection of unwanted features were derived and developed for some video processing applications. There are various types of feature detectors like edge, corner and points which can be detected from a video frame. To detect those features, feature point detectors like SIFT, ORB, FAST, BRIEF and FFME are more reliable with good efficiency, less memory size and high processing speed. These feature point detectors are compared and evaluated under parameters of speed and number of feature point matches. Although, FFME algorithm has high speed when compared to other feature point detection algorithms, it is prone to more errors caused by illumination differences, as it detects the feature points using Gradient magnitude of the pixel. To overcome this, we have introduced a novel work that uses FFME with gradient phase technique. The resultant output exhibits high processing speed with less error when compared to other feature point detectors and even FFME with gradient magnitude.

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1. Introduction

In era of automation, the computer vision and image analysis impact a wide range of applications such as object and pattern recognition, segmentation and scene inspections. In recent years, researchers have applied these fields to many application areas from robotic navigation to surveillance to autonomous supervision [1]. Due to the cheaper hardware components and less processing power, computer vision based applications have significant role in intelligent autonomous systems to make decisions depending on what they see, like images and videos obtained by an image sensor, an integral part of the imaging system [2].

In certain conditions, the purpose of this work is to compare various feature detectors used in video processing. The basic principles of feature point detectors employ feature detection techniques like SIFT (Scale Invariant Feature Transform), FAST (Features from Accelerated Segment Test), BRIEF (Binary Robust Independent Elementary Features), ORB (Oriented FAST and Rotated BRIEF) and FFME (Fast Feature Based Motion Estimation). These techniques were tested in this work and their performance metrics were estimated in terms of speed with respect to fps and number of feature points detected. Finally, for improving the detection of feature points in videos with illumination changes the FFME algorithm was modified with gradient phase calculation and the results were studied in detail.

In general, the aim of this paper established the motion estimation. It is the process of determining the motion between two or more than two frames in a cascaded image. Motion compensation is the technique of predicting and reconstructing a frame using a given reference frame and a set of motion parameters. It sometimes called as motion determination. Motion estimation is not only used in the field of video compression but also in the field of region of interest detection, segmentation, de-interlacing, and object tracking and so on. Feature based motion estimation and compensation have traditionally been performed, using feature detecting techniques. They used to quick response, simple to implement and fairly effective over a wide range of video [3]. To predict the frames, feature-based motion estimation is more effective and practical. It separates frame by frame and finds out the displacement of the best-matched feature points from previous frame as the motion vector to the same feature in the current frame within a search window. Based on feature matching or other matching criteria, the displacement of the best matched feature will be described as the estimated motion to the feature in the current frame.

The best open source platform to implement the computer vision has long been the standard platform, which is called OpenCV (Open source Computer Vision) [4] and is very much recommended for real-time video processing.

2. Related work

In accordance with N. Gupta and L. Kanal, a 3D motion estimation of objects from motion field was proposed. They derived a system for estimating the parameters of motions from linear equations. But, it could be applicable only for the smooth surfaces [5]. Gandhi et al. developed a detection and tracking of significant features from obstacles on aircraft runway. They implemented the separation work of moving and static objects by applying thresholding algorithms. They applied the SUSAN corner detector for detecting the features. They didn't estimate the feature points for various sequences of image at different situations [6]. In 2003, Clark F. Olson et al. implemented robot navigation with the help of stereo ego-motions. They studied the performance of ego-motions using maxima-likelihood algorithm. The approximation of error was 1.2% [7]. Yamaguchi et al. approached the estimation of vehicle ego-motions for detecting moving objects by using monocular camera. Ego-motions were estimated with the help of the 8-point and RANSAC algorithms. Then, the moving object regions were calibrated by the magnitude of the cameras optical flow. The road regions were detected using least-mean square estimators. They achieved higher accuracy of detected regions although not studied about stationary objects [8]. Schmudderich et al. investigated the motion of objects using optical flow and kinematics. They introduced the term ego-motion flow (EMF) from the combination of ego-motion and optical flow. The segmentation of moving objects was performed by using optical flow and kinematics [9]. Shen et al. suggested for video stabilization using PCA and SIFT algorithm to estimate the motion features using SIFT algorithm and extracted by using RANSAC algorithm. Further, the combination of SIFT-BMSE function was proposed to get accuracy, even though the computational speed was found to be slow [10]. O. Kwon and Y. Ha applied SIFT algorithm for the panoramic videos. They approached SIFT algorithm based color-invariant features image stitching for reducing the illumination. They didn't measure the

accuracy of the proposed method [11]. Xu et al. attempted for video stabilization based on fast feature detectors. They applied ORB, SIFT and SURF algorithms for global motion estimations. Among them, ORB exhibited more stability when motion smoothing Gaussian filter was applied. The processing speed was limited because of more number of image processing steps involved to detect features [12]. Liang et. al., applied SIFT and NSCT based spectral pattern matching problem. The matching performance and accuracy were evaluated for 0 to 20th frames in video. The drawback of this work was when the frame size was increased then the accuracy was found to decrease [13]. M. Angelopoulou and C. Bouganis proposed the real time ego-motion estimation system of an unmanned vehicle on FPGA platform. Features were detected by block matching algorithm. But, it required more memory and high processing time to estimate the features [14]. Droeschel et al. designed ego-motion estimation using time of flight cameras for robots. The optical flow features were matched by fusion of SIFT and IMU algorithms. It showed more camera specific errors and low processing speed [15]. Recently, the research contributions in ego-motion estimation has been attracted various applications like 3D reconstruction, background subtraction for moving cameras and camera pose estimation in self-driving [16-18].

3. Proposed Work

The proposed work consists of the following stages to obtain a reliable singular feature point detector. Fig.1 shows the overall proposed system architecture for the detection of reliable feature points.

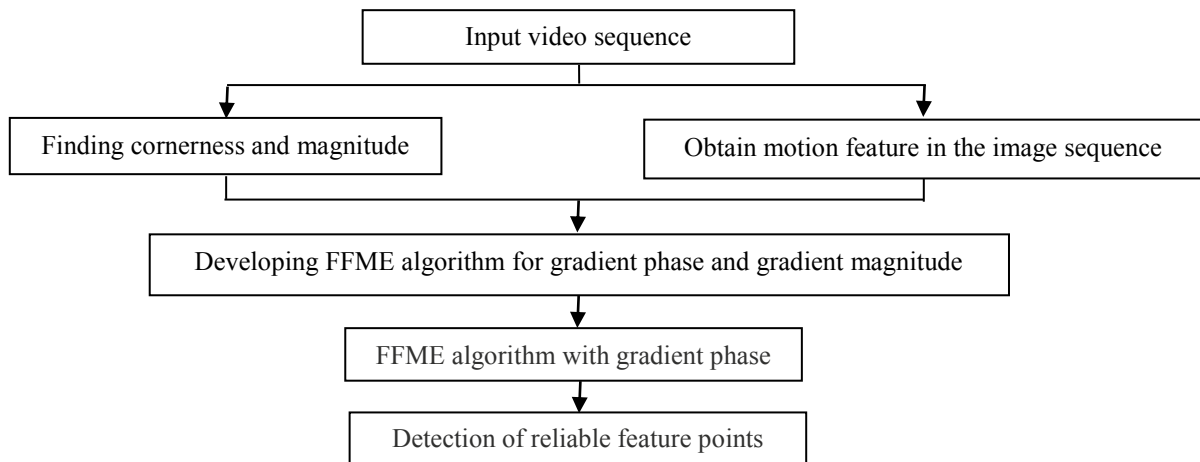


Fig. 1 Proposed system architecture

3.1. Feature detectors

Feature or attribute of an image is chosen as information for processing and computing a task in the field of computer vision. Features may be any sort of specific structures in the image such as edges, points or objects. Features, by definition, are locations in the image that are perceptually interesting in an image. Features are used for image alignment, object recognition, 3D reconstruction, motion tracking, indexing and content-based retrieval and robot navigation. Image feature detection is helpful to solve the problems on object recognition, motion correspondence and image registration [19]. Previous works on feature detection include application of statistical methods for the estimating the position of corners and edges. These methods could be used in several image processing applications like aerial image analysis, analysis of airport images and map matching. A robust algorithm for feature detection has the identification characteristic, which is invariant with respect to translation, rotation, scaling and deformation functions [20]. But, evolving autonomous systems need more precise features of an image other than edges and corners alone. For objects, image features like corners and curvatures of locations could be processed and analyzed, whereas nose, mouth and eyes and shape of the face could be used for feature extraction.

The edge and corner are not the only general features for feature detection, there are many other features reported in the literature. The method, we describe in this work is both robust and of general utility, and has been tested successfully on several real time feature detection. The main focus of this paper is to demonstrate the robustness of the proposed fast key feature detection mechanism with gradient phase to detect the feature as we explore its applications to several video processing problems.

In demonstrating the utility of this Fast Feature based Motion Estimation (FFME) feature detection technique with phase and magnitude, we have applied and analyzed their performance in feature key-point detection in live video sequences.

3.2. Feature Detection and Extraction

The repeatability is the desirable property in a feature detector, since two or more different images of the same scene will not be detected as the feature. Feature detection is a primary level image processing technique. The preliminary operations on a frame or picture consider every pixel to verify for a region of interest as feature point. Furthermore, when the feature detection contains time constraints and expensive to computation, then high level algorithms are used to detect the feature. Then only some parts of the image will be identified as features.

Feature could be extracted when any features have been detected by applying a local image patch around the feature. This process output is called feature descriptor or feature vector [21].

3.3. FFME

FFME (Fast Feature Based Motion Estimation) is an algorithm specifically designed for video stabilization. Without the requirement of specific hardware, sparse motion vector field can be computed in real-time using the proposed FFME algorithm. In order to perform complex mathematical operations involved in real-time processing, the computation of motion vectors is confined to the subset of image points. Further, high quality sparse motion vector fields can be yielded to achieve robustness in motion estimation technique by taking into account the various noise and artifacts present in the 3D image.

In FFME, the corners and magnitude of the gradients are extracted as the key singular points in the present frame of the video sequence. Then, the neighbourhood of every singular point describes the main qualities by using a complex descriptor vector, which improves illumination and orientation related changes in 3D camera movements and thus ascertaining its robustness against artifacts. Based on the past descriptor vector of maximizing a similarity is measured in “ n ” consecutive different singular points in images. The set of relative singular points are used to determine the sparse motion vector field representing the image motion outline [22]. The extracted feature points are then used in the second stage to obtain motion of feature in the image sequence. From the Eq.1, the Matching Score (MS) is estimated. Here, the motion singular points belong to the singular points from influence function to n^{th} frame.

$$MS(\overrightarrow{msp_i} \in sp_{fin}^n) = \min_{\overrightarrow{msp_i} \in sp_{pj}} \min_{n+1} \left\| D_{fin}(\vec{p_i}) - D_{fin}(\vec{p_j}) \right\| \quad (1)$$

3.4. FFME algorithm with gradient phase

Using Rayleigh distribution, the phase was calculated from the derivation of magnitude. From Eq.2, the gradient phase was derived from the magnitude. Here, RD represents the Rayleigh distribution and the image gradient ∇X is derived for each pixel array m, n .

$$RD(\|\nabla X(m,n)\|) = \frac{\|\nabla X(m,n)\|}{\sigma^2} e^{-\frac{\|\nabla X(m,n)\|^2}{2\sigma^2}} \quad (2)$$

4. Result and discussion

4.1. FFME algorithm output

FFME algorithm is used for finding reduced number of singular points. Generally, it was used for video stabilization applications. But, using the correspondence acquired from feature points extracted from the singular points, it is used in feature extraction in an efficient way. This technique allows the user for extracting required feature in less processing time. The algorithm was tested in a video sequence and the output is shown in Fig. 2 with changing in illumination.

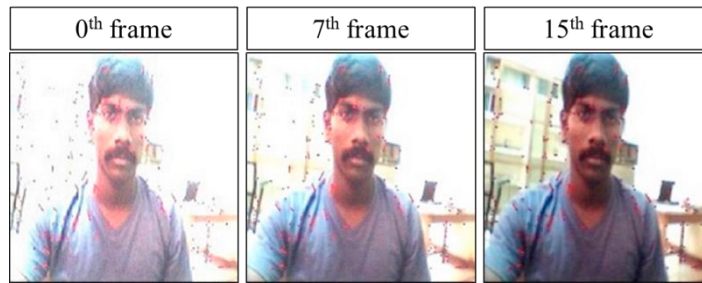


Fig. 2. Output of FFME with magnitude phase algorithm

4.2. Feature Detection and Extraction

Gradient of an image is used to extract some common features using convolutions. The magnitude and intensity change in original image in a given direction can be estimated from the information of each pixel of a gradient image. The gradient phase of an image gives the orientation of the selected pixel in X and Y directions. Thus, by using this gradient phase in FFME, the errors caused due to lighting and camera changes are reduced. The algorithm was developed and implemented in a sequence of video frames and the output is shown in Fig. 3 with changing in illumination.



Fig. 3. Output of FFME with gradient phase algorithm

4.3. Comparison of feature detectors

The feature detectors employed in this study and their respective outputs are given in Fig. 4. The performance comparison of the feature detectors viz., FFME with gradient phase and FFME with gradient magnitude are given in Table 1 and Fig. 4, respectively. From the Fig. 4 and Table 1, it was inferred that the FFME with gradient phased

technique was found to detect maximum number of feature points at a higher frame rate, whereas the other methods scored maximum matching scores at a very slower frame rate. As a result, the FFME with gradient phase was found to be a better method than the others.

Table 1. COMPARISON OF FEATURE POINT DETECTORS

Feature detectors	Frame rate (fps)	Speed (s)	Matching scores
FFME (magnitude phase)	5.9	166	201
FFME (gradient phase)	7.1	139	203

4.4. Comparison chart for FFME with gradient magnitude and gradient phase

The FFME and FFME with gradient phase algorithms were tested with a same video sequence for comparison of their performance. The calculated speed of execution of the respective algorithms and their matching scores are graphically represented as shown in Fig.4. From the graph, it was observed that the speed of the execution is found to increase gradually with respect to matching score for the FFME than the FFME alone. As a result, the number of reliable feature points was found to be efficiently detected using gradient phase based FFME.

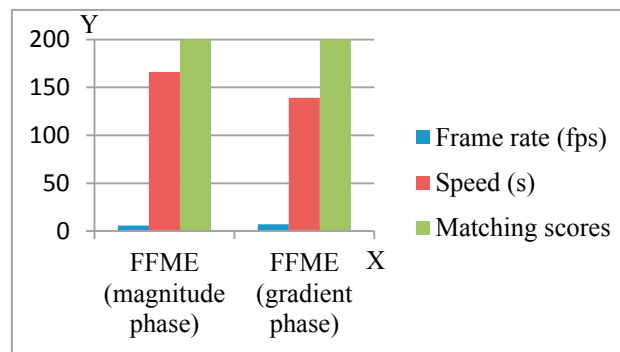


Fig.4. Comparison chart of FFME with gradient magnitude and gradient phase

5. Result and conclusion

In this work, a comparative study of two different feature detectors was carried out in terms of speed of the video frames and number of matching points of the input video on OpenCV environment. Among the compared algorithms, FFME based detection was chosen to boost up the detection of reliable feature points because of its outstanding performance in speed and number crunching capability. To rectify the errors caused due to illumination variation in real-time live video sequences, gradient phase based feature detection was incorporated with FFME algorithm. The results showed an increase in number of detected feature points with reliable singularity and also a decrease in processing time. The total frames taken for the evaluation is 15 fps. Finally, the obtained feature points can be further used for motion estimation techniques in computer vision based applications.

6. Future work

The simulation work doesn't meet the required time as well as speed in the real-time. So, we will implement the proposed algorithm in the hardware environment like FPGA (Field Programmable Gate Array), since it contains needed peripherals and high accurate global clock on a chip. This will meet the required time as well as speed in the future endeavours.

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